Automatic Vehicle License Plate Recognition System based on Image Processing and Template Matching Approach

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***Abstract*—A vehicle license plate recognition system is an important proficiency that could be used for identification of engine vehicle all over the earth. It is valuable in numerous applications such as entrance admission, security, parking control, road traffic control, and speed control. However, the system only manages to identify the license number and needs an operator to control the collected data. Therefore, this paper proposes an automatic license plate recognition system by using the image processing and template matching approach. The current study aims to increase the efficiency of license plate recognition system for Universiti Malaysia Perlis (UniMAP) smart university. This venture comprises of simulation program to recognize license plate characters where a captured image of vehicles will be the input. Then, these images will be processed using several image processing techniques and optical character recognition method in order to recognize the segmented number plate. The image processing techniques consist of colour conversion, image segmentation using Otsu's thresholding, noise removal, image subtraction, image cropping and bounding box feature. The optical character recognition based on template matching approach is used to analyse the printed characters on the segmented license plate image and to produce an output data consisting of characters. Overall, the proposed automatic vehicle license plate recognition system is capable to perform the recognition process by successfully recognizing license plate of 13 cars, from a total of 14 cars.**

Keywords—license plate recognition, vision system, image processing, image segmentation, optical character recognition

I. INTRODUCTION

Malaysia is a vast growing country. The nation is undergoing magnificent transformation in its economy and population. As population increases, the stream of traffic on the roads also increases. Due to the following factor, crime related to traffic increases at a greater rate. Various cases of burglary, hit and run, grabbing and on-street fatalities, stay unsolved on the grounds that the vehicle included could not be perceived precisely. This is because the human eye is unable to confirm the characters of license plate from vehicles that move fast [1].

In this day and age, monitoring vehicles for law enforcement and security purposes is a difficult problem due to the number of automobiles on the road [2]. One of the examples lies for security system in university campus gate entrance, as it is time consuming for the security guard to visually and physically check the license plate of every vehicle. Additionally, it is not feasible to employ a number of guards to act as full-time license plate inspectors. Right off the bat, the utilization of stickers as a technique for recognizable proof to grant entry to the campus is not fool proof. It has security flaws, for example, simplicity of duplication. There should be a way to track and identify license plates without continuous use of human services [3]. This can be achieved by using automatic number plate recognition (ANPR) which uses image processing technique and optical character recognition (OCR) approach to define the individual character of the number plate.

Thereby, several previous studies have been researched for vehicle license plate recognition system. There are various methods used in vehicle license plate recognition system in order to obtain the data of the license plate [4-6]. For instance, Rajput and Som [7] proposed an automated vehicle license plate recognition system framework that automatically seeks to overcome the shortcomings of existing approaches. It consists of three phases: image capture, license plate localization, and number recognition. This approach is based on a single-stage wavelet transform, and has proven to be effectived in many situations and is therefore useful for real applications. Experimental results indicate that the proposed method has achieved high accuracy with average accuracy of 97.33%.

Quiros *et al.* [8] proposed automated plate detection for the use of intelligent transport system by using image processing techniques such as edge detection and contour matching. The proposed method has achieved an average accuracy of 96.67%. A license plate detection based on colour detection method along with channel scale space technique has been proposed by Davix *et al.* [9]. The proposed method calculates the aspect ratio and the license plate is distinguished to identify the vehicles. The CIE-XYZ colour model is used for the process and produced an accuracy of 94.23%.

Mahamad *et al.* [10] proposed a simplified Malaysian vehicle plate number recognition. The techniques that have

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been used are digital image processing and OCR method. The proposed method does not give any information about the accuracy performance. Xie *et al.* [11] proposed a CNN-based MD-YOLO framework for multi-directional car license plate detection. The proposed method used precise rotation angle prediction and a fast intersection-over-union evaluation strategy. The experimental results show that the proposed method outclasses over other current methods in terms of better accuracy and lower computational cost. Mukherjee *et al.* [12] proposed a robust algorithm for the extraction of image morphology, spatial filtering and mapping features and characters that are used for vehicle number plate recognition. The proposed method has achieved an average accuracy of 79.30%.

As digital image processing and OCR play a major role in license plate recognition, the current study will utilize the potential of image processing techniques and OCR method for automatic vehicle license plate recognition system. The proposed system aims to increase the efficiency and reliability of security management system in UniMAP.

II. Methodology

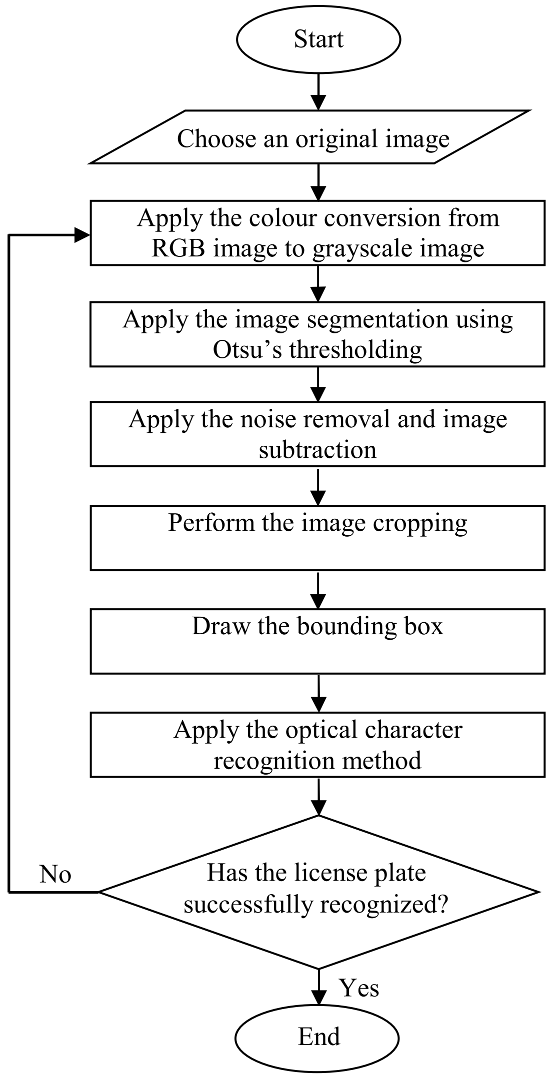


Fig. 1. The image processing and license plate recognition steps for the proposed system.

In this paper, there are several image processing techniques will be applied in order to obtain the segmented area of interest of license plate images, as well as to recognize the characters of the license plate. In this study, the area of interest is referred to the segmented license plate. The procedures to develop the image processing and license plate recognition steps for the proposed system are illustrated by the flow chart in Fig. 1.

A. Data Acquisition

Data acquisition is an important and critical procedure for this project. The key reason is the image quality certainly defines the outcome of the recognition result. The data acquisition procedure is done by acquiring images of vehicles. In this project, images of a total of 14 cars have been captured for evaluation process. The images are captured using an iPhone 5s smartphone. The camera specification for an iPhone 5s is 8mp for the rear/main camera. As this research is conducted for initial study, an iPhone is used because it produces an image with high quality. The captured images were saved in jpeg (*.jpg) format. This is the default format for the images captured in an iPhone.

Fig. 2 shows several images of cars. Based on these images, the images vary according to the distance and angle. Different angles and distances from the license plate and camera have been collected. The images have been captured in the morning, evening and night. In addition to it, vehicles with various colours have been chosen for this research.

(a) Car A (b) Car B (c) Car C

Fig. 2. The sample images of the captured vehicle.

B. Colour Conversion Technique

The iPhone 5s smartphone produces RGB image. The complexity of the code and difficulty in visualisation is the main reason RGB image is converted into grayscale image. In addition, the RGB image is also difficult to be segmented. Thus, the captured images are converted from RGB image to grayscale image to produce an essential image for the Otsu's thresholding segmentation procedure. This conversion is required to reduce the complexity of an image. Moreover, a good segmentation result based on grayscale image will be produced as compared to the RGB image.

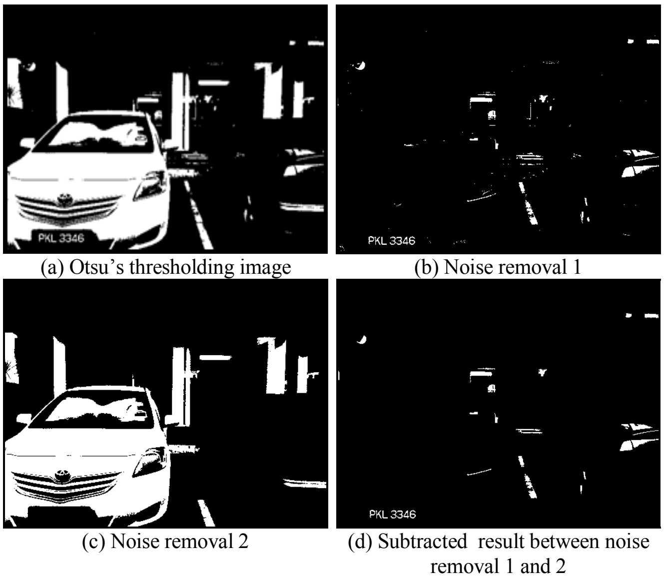
C. Image Segmentation using Otsu's Thresholding.

The next step is a crucial step in image processing technique. Image segmentation is applied to implicate the image into a more specific distribution of pixel. Thus, the converted grayscale image will be processed using Otsu's thresholding technique. Otsu's thresholding is a method used in image segmentation to automatically calculate a threshold value using an iterative method. It also controls the ideal threshold value by maximizing the overall variance between the "background" and "object". The image produced after

Otsu's thresholding would be easier to process in order to produce a segmented license plate image.

D. Noise Removal and Image Subtraction

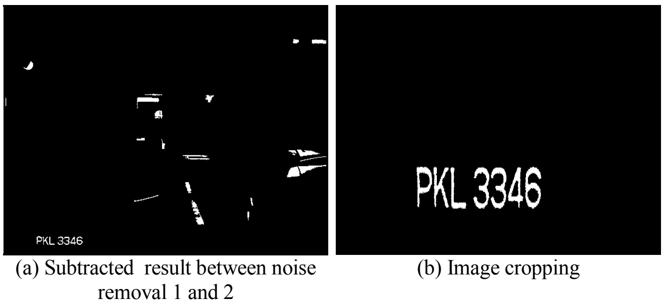
The next step is the noise removal and image subtraction. This step is important in order to remove the small spots and undesired objects in the image. These small spots and undesired objects will be removed to increase the accuracy of recognition process. At first, these undesired objects are removed by using the region growing technique. By using this technique, any undesired objects below the stated pixels would be removed. This process is then supported by the image subtraction process. The key usefulness of subtraction is the enhancement of difference between images which result a desired image for segmenting the license plate. Fig. 3 shows sample results of noise removal and image subtraction processes.



(a) Otsu's thresholding image (b) Noise removal 1

(c) Noise removal 2 (d) Subtracted result between noise removal 1 and 2

Fig. 3. The result image of noise removal and image subtraction.



(a) Subtracted result between noise removal 1 and 2 (b) Image cropping

Fig. 4. The segmented license plate image resulting from image cropping technique.

E. Image Cropping for the Segmented License Plate

The next step is the image cropping. This process is perform in order to crop the segmented license plate by using the standardize values. The image cropping is done according to the desired angle and distance between the license plate and camera of the input image. This step is important to increase the accuracy of the recognition process. The output image of this process is a segmented license plate image. Fig. 4 shows a sample segmented license plate image resulting from image cropping technique.

F. Bounding Box

The segmented license plate image is the result of performing all the image processing techniques. Thus, the implication of recognition processes start with identifying the alphabet and number from the background using bounding box feature. The bounding box is created on the image where the system detects the presence of object. This certainly proves the isolation process between object and background is very important.

G. Optical Character Recognition based on Template Matching Method

OCR is a process to identify the data from the segmented license plate. This process used to recognize printed information on an image to produce an output data consisting of characters. The process is done by analysing the bounding box on the images and create algorithm to compare the similarity of the obtained character with the training data. This algorithm is also known as template matching. Template matching is often known as the basic core for OCR. This method has been used widely in recognition process. Fig. 5 shows the training data used for the template matching process.

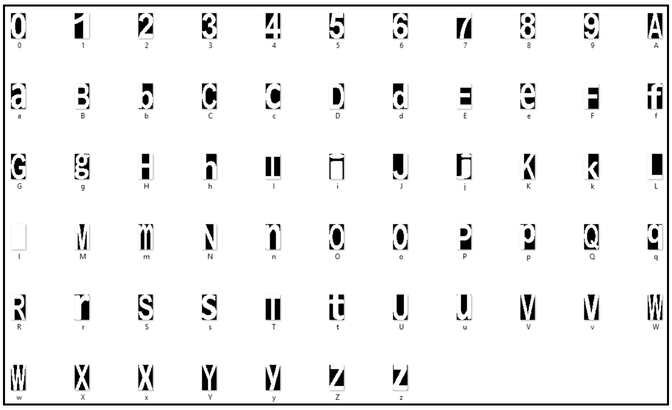


Fig. 5. OCR template matching test data.

III. Result And Discussion

This section provides the results that have been obtained based on application of image processing techniques and OCR in order to recognize the vehicle license plate. The detail of findings which have been yield through the implementation of this research will be discussed in this section.

A. Results of Image Processing Techniques and Optical Character Recognition on Three Different Cars

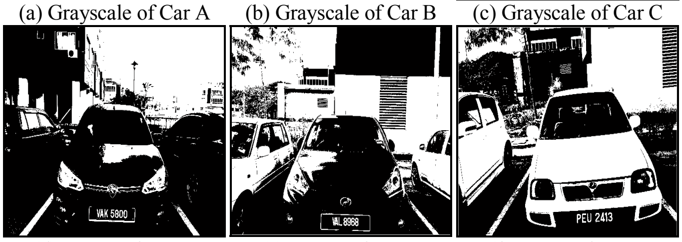
This section provides the results after applying the proposed method on three different cars that have been selected. These cars are named as Car A, Car B and Car C as shown in Fig. 6. By referring to these samples images, the car images have been captured with different angles from the license plate and the camera. In addition to it, vehicles with light and darker colours have been used in order to test the capability of the proposed image processing procedures in license plate segmentation process.

(a) Car A (b) Car B (c) Car C

Fig. 6. The original car images.

As the original car images are in RGB image, it would be difficult to segment the license plate as the segmentation process needs the consideration of the three R, G and B values. Therefore, it is always easier to convert an RGB image to a grayscale image and then into a binary image to facilitate the image segmentation process. During this process, Otsu's thresholding has been applied in order to segment the characters of license plate from background region. The result of Otsu's thresholding will produce a binary image as shown in Fig. 7.

(a) Grayscale of Car A (b) Grayscale of Car B (c) Grayscale of Car C



(d) Otsu's of Car A (e) Otsu's of Car B (f) Otsu's of Car C

Fig. 7. The car images after undergoing grayscale conversion and Otsu's thresholding technique.

By referring to Fig. 7, it can be clearly seen that the characters of the license plate have been properly segmented from the license plate background. By focusing on the license plate, the characters appear in white colour, while the license plate background appears in black. However, the OCR could no directly be performed on this resultant image, as the background of the car will reduce the accuracy of the recognition process. Hence, the thresholding process is then followed by execution of noise removal by using region growing technique and image subtraction process on the binary image. By applying these two processes, any undesired objects in the image which is referred to the background of the car have been removed to ensure the characters on the license plate are properly maintained.

Later, image cropping process has been perform in order to crop the segmented license plate image by using the standardize values. This process is performed to ensure the output image only consist the characters of license plate, hence increasing the accuracy of character recognition process. In order to perform the character recognition process, the bounding box has been created on each character of the segmented license plate. By using the bounding box feature, the extraction of the characters on the license plate will only requires the bounding location of each object in the selected candidate region. Then, for each extracted bounding box, objects with a maximum area are maintained.

Finally, the character recognition process based on template matching approach is used to recognize the printed information on an image to produce an output data consisting of characters. Figs. 8 and 9 show the results for the three different cars after applying the bounding box feature and OCR using template matching approach, respectively.

(a) Car A (b) Car B (c) Car C

Fig. 8. Results of images of segmented license plate after performing the image cropping and bounding box process.

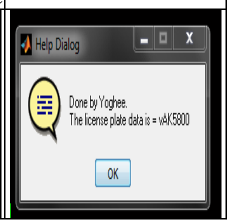
(a) Car A (b) Car B (c) Car C

Fig. 9. Results of images of segmented license plate undergoing Optical Character Recognition using bounding box feature.

Table I, II and III represent the additional results of image processing and recognition process for Car A, Car B and Car C, respectively. These results contain further analysis based on different distances between the car license plate and the position of the camera. These distances are measured at 2.0m, 2.3m and 2.5m. The height of the camera position was fixed at 145 cm. This height has been suitable in obtaining the full image of the license plate from various angles.

Based on the results that have been obtained, all the images that have been captured from different angles and distance have been successfully processed and recognized. The camera angle from the right and the distance 2.0m between the license plate and the camera has been chosen for the most suitable placement of the camera. This is due to the ideal position for a camera in a gate entrance is at the right side. The distance 2.0m has been chosen because the image quality is better at a nearer position and it covers perfectly the whole car.

TABLE I. THE RESULTS OF IMAGE PROCESSING AND RECOGNITION PROCESS OF CAR A

Front	Otsu's Thresholding Image	Segmented Image of the License Plate	Recognition Result
2.0m			

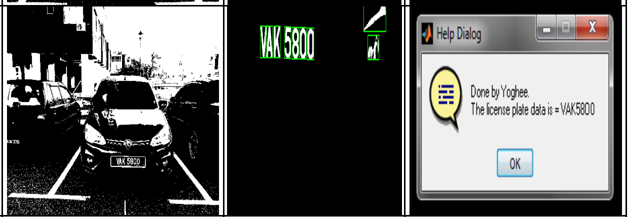
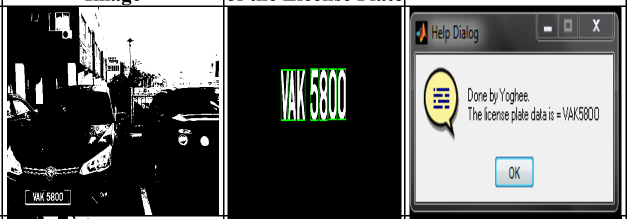
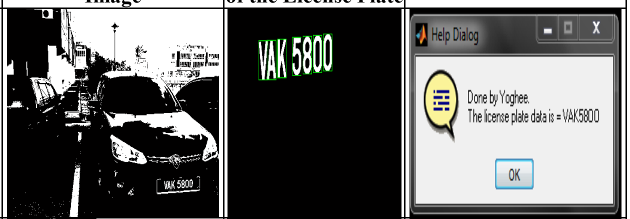
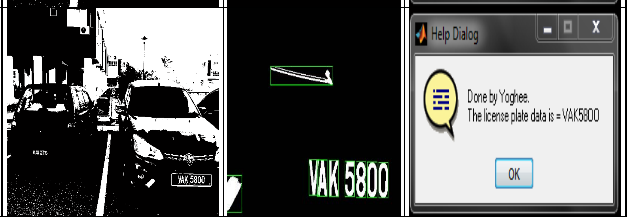
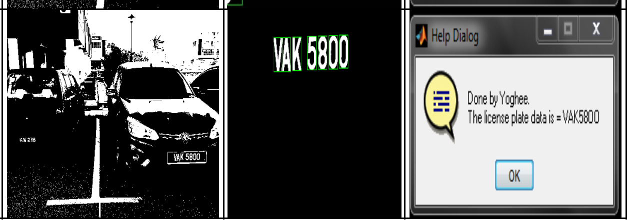
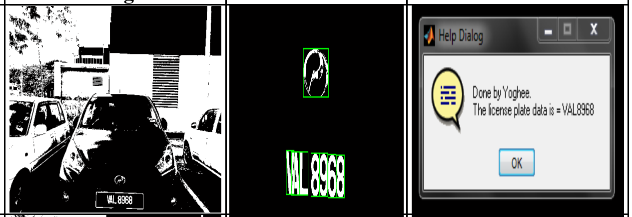
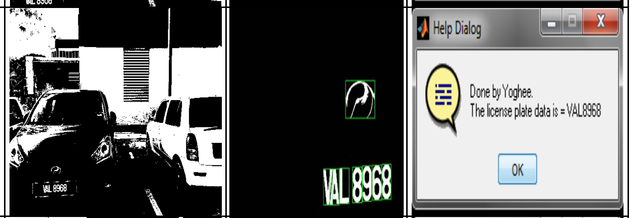
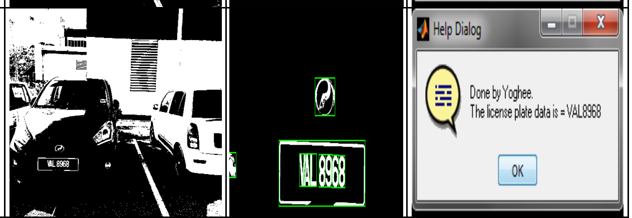
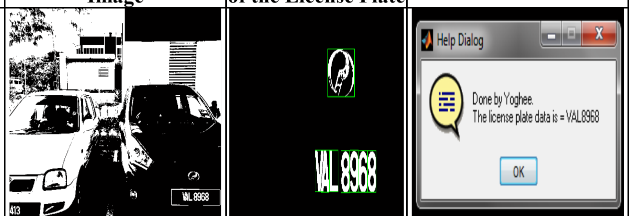
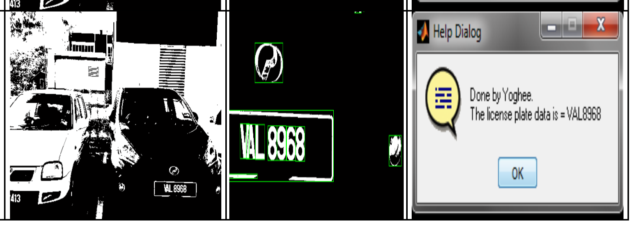
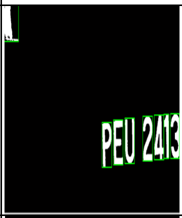
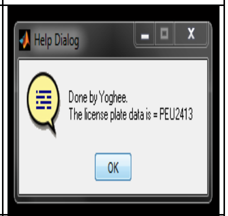
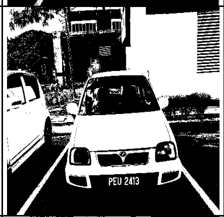
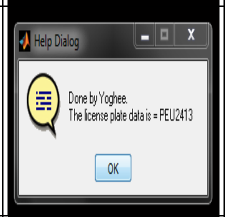
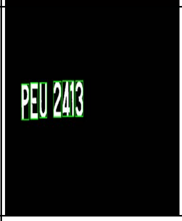
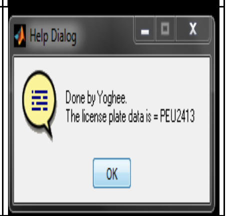
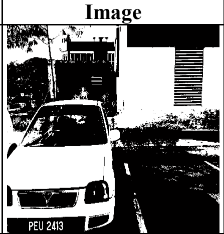
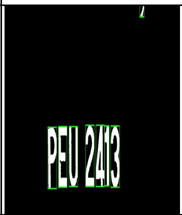
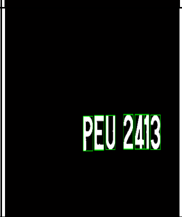
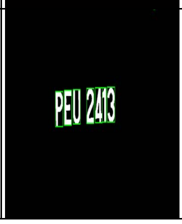
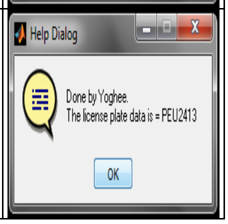
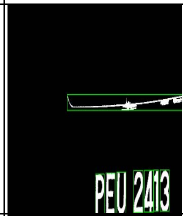
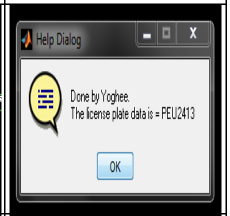
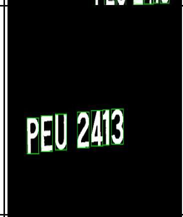
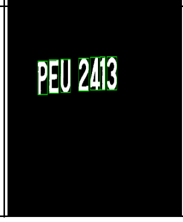
2.3m			
2.5m			
Right	**Otsu's Thresholding Image**	**Segmented Image of the License Plate**	**Recognition Result**
2.0m			
2.3m			
2.5m			
Left	**Otsu's Thresholding Image**	**Segmented Image of the License Plate**	**Recognition Result**
2.0m			
2.3m			
2.5m			

TABLE II. THE RESULTS OF IMAGE PROCESSING AND RECOGNITION PROCESS OF CAR B

Front	**Otsu's Thresholding Image**	**Segmented Image of the License Plate**	**Recognition Result**
2.0m			
2.3m			
2.5m			
Right	**Otsu's Thresholding Image**	**Segmented Image of the License Plate**	**Recognition Result**
2.0m			
2.3m			
2.5m			
Left	**Otsu's Thresholding Image**	**Segmented Image of the License Plate**	**Recognition Result**
2.0m			
2.3m			

2.5m			

TABLE III. THE RESULTS OF IMAGE PROCESSING AND RECOGNITION PROCESS OF CAR C

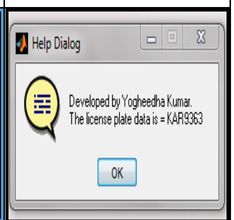
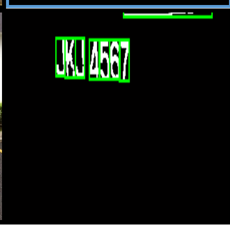
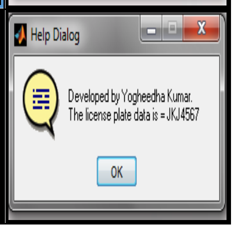
Front	Otsu's Thresholding Image	Segmented Image of the License Plate	Recognition Result
2.0m			
2.3m			
2.5m			
Right	**Otsu's Thresholding Image**	**Segmented Image of the License Plate**	**Recognition Result**
2.0m			
2.3m			
2.5m			
Left	**Otsu's Thresholding Image**	**Segmented Image of the License Plate**	**Recognition Result**
2.0m			
2.3m			
2.5m			

B. *Results of Image Processing Techniques and Optical Character Recognition on 11 Different Cars*

The performance of the proposed license plate recognition system has been furthered evaluated by analysing an additional of 11 cars license plate images and the results are represented in Table IV. The image acquisition for this analysis has been performed throughout three phases which is during the morning, evening and night. The camera angle from the right and the distance 2.0m between the license plate and the camera has been standardize for the most suitable placement of the camera.

By analysing the results, a total of 11 cars have been evaluated in this process and 10 cars have been successfully recognized. The reason of one missing car (DCF2906) is due to the low image quality and the number "2" on this car has the shape of the alphabet "Z". This may cause a disruption in the accuracy of the reading. Overall, the results show that different time of a day does not really affect the accuracy and the recognition process could work in minimal lighting.

TABLE IV. THE RESULTS OF IMAGE PROCESSING AND RECOGNITION FOR 11 DIFFERENT CARS

Original Image	Segmented Image of the License Plate	Recognition Result
		
		

All the results and the insights have been processed and discussed in detail. In this research various conditions such as distance, angle and phases of the day have been tested and evaluated. Hence, the camera angle from the right and the distance 2.0m between the license plate and the camera has been chosen for the most suitable position. This is due to the ideal position for a camera in a gate entrance is at the right side. The distance 2.0m has been chosen because the image quality is better at a nearer position and it covers perfectly the whole car. Besides that, the recognition process could work in minimal lighting so the different time of the day doesn't affect the accuracy.

IV. Conclusion

In this paper, a total of 14 cars have been evaluated. The proposed recognition system is specialized for standard license plate characters only. Hence, the only mismatch happened in this system when the car tested had a non-standard license plate characters. Based on the results that have been obtained, it is found that the proposed system is able to produce good recognition results even though the car images are captured under minimal light condition. In addition to it, variation of colours of the car does not affect the recognition process. Hence, the proposed automatic vehicle license plate recognition system is capable to perform the recognition process by successfully recognizing license plate of 13 cars, from a total of 14 cars. Thus, the results significantly demonstrate the suitability of the combination of image processing techniques and OCR based on template matching approach in vehicle license plate recognition system.

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